(a) Truth table

x_1	x_2	x_3	x_4	$f(x)$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Binary output column

f = 0101010101010111

Integer encoding

$Int(f) = (0101010101010111)_2$

$= 21847$

(b) Boolean expression

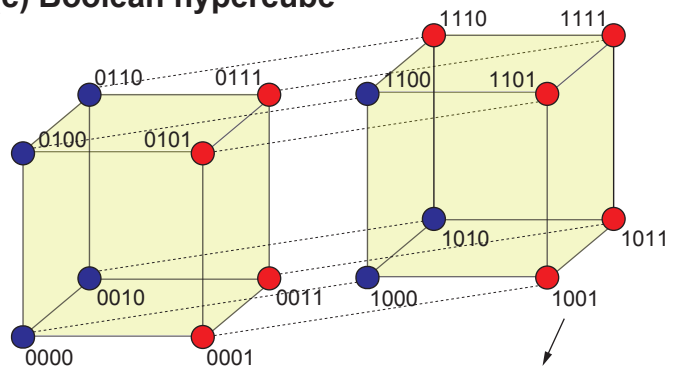
$$f(x) = x_4 \vee (x_1 \wedge x_2 \wedge x_3)$$

where

$\vee$: logical 'OR' operator

$\wedge$: logical 'AND' operator

(c) Boolean hypercube



: 1 ('on') : 0 ('off')

Order of variables in input labels

$x_1 \; x_2 \; x_3 \; x_4$